Machine Learning-Based Classification of Antimicrobial Peptides and toxicity prediction (a comparitive study)

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INTRODUCTION

- Antimicrobial peptides (AMPs) are promising alternatives to conventional antibiotics
- Experimental identification of AMP activity and toxicity is expensive and time-consuming
- Machine learning provides a fast and scalable computational approach

Literature Review

Title	Dataset	Methodology	Merits	Demerit
Deep Learning in Antimicrobial Peptide Prediction, 2025	XUAMP dataset	DL model called iAMP-CRA, which utilizes a Convolutional Recurrent Neural Network (CRNN) with Self-Attention	BiLSTM had an accuracy of 90.80 %, had negative samples	Data Imbalance, developed to target only a single microorganism (Gram-negative E. coli)
MSCMamba: Prediction of Antimicrobial Peptide Activity Values by Fusing Multiscale Convolution with Mamba Module, 2025	APD3, DRAMP, DADP, and YADAMP	Regression model called MSCMamba	Accuracy was better than existing models (10.6 % increase)	Computationally complex, does not predict well on real-world unseen data
deep-AMPpred: A Deep Learning Method for Identifying Antimicrobial Peptides and Their Functional Activities, 2025	APD3, dbAMP, LAMP, and DRAMP	• A binary classifier that distinguishes AMPs from non-AMP sequences. • A multilabel classifier that predicts the specific functional activities. • CNN AND BiLSTM	Binary classifier for AMP recognition and a multi-label classifier for acitivity classification	Multilabel Imbalance, only depends on sequence information

Literature Review

Title	Dataset	**Methodology**	Merits	**Demerit**
CalcAMP: A New Machine Learning Model for the Accurate Prediction of Antimicrobial Activity of Peptides, 2023	CAMP, DBAASP, DRAMP, LAMP2, and YADAMP.dataset	Ensemble of Machine Learning Models - Random Forest, CatBoost, LightGBM, and XGBoost.	Included quantitative prediction of the antimicrobial activity (MIC) of synthetic peptides	Lack of Species-Specific Data, To ensure accuracy, large amount of experimental data was discarded.
sAMPpred-GAT: Prediction of Antimicrobial Peptide by Graph Attention Network and Predicted Peptide Structure, 2023	XUAMP, APD3, DRAMP, LAMP, CAMP, dbAMP, YADAMP	Incorporates 3D predicted structure information (via AlphaFold) into a Graph Attention Network	Graph Attention Network (GAT) included high-level information about the folds of the peptides	Only focuses on the binary classification of "AMP vs. non-AMP" , very low accuracy of 68.7 %

Problem Statement

Large AMP databases contain numerous activity types with severe class imbalance, making simultaneous prediction of activity and toxicity challenging; moreover, most existing approaches rely on a single prediction task or algorithm, highlighting the need for a comparative machine learning study for reliable AMP classification.

Objectives

- To preprocess dataset to handle class imbalance
- To classify antimicrobial peptides into major activity classes
- To predict peptide toxicity using computational methods
- To compare the performance of multiple machine learning models
- Compare model performance between physicochemical features and ESM-2 based transformer embeddings
- Explainable-AI : with SHAP

Dataset and Preprocessing

	A	B	C	D
1	ID	Sequence	Activity	
2	CAMPSQ1	KGIRGYKG	Antifungal	
3	CAMPSQ1	LQPGAPNF	Antifungal	
4	CAMPSQ1	LQPGAPKL	Antifungal	
5	CAMPSQ1	SVIGCWTK	Antibacterial	
6	CAMPSQ1	IGCWTKSI	Antibacterial	

Activity	Training Set (Before SMOTE)	Training Set (After SMOTE)
Antibacterial	14,539	14,539
Antifungal	1,735	14,539
Antiviral	1,039	14,539
Total Training	17,313	43,617

- Data Source: Curated from DRAMP, DBAASP, CAMP3, GRAMPA
- Class Distribution: Antibacterial, Antiviral, Antifungal.
- Contains 26k peptide sequences
- Removed unwanted activity labels
- Used SMOTE for handling class imbalance

Feature Extraction

Feature	Calculation Method	Biological Activity
Length	len(seq)	Structural complexity
Hydrophobicity	count(A,I,L,M,F,W,Y,V) / Length	Membrane penetration
Positive Ratio	count(K,R,H) / Length	Bacterial surface binding
Negative Ratio	count(D,E) / Length	Electrostatic repulsion
Cysteine Ratio	count(C) / Length	Structural stability (Disulfide bonds)
Net Charge	(Positive - Negative) / Length	Total electrostatic potential

1. Physicochemical Descriptors (6 Features)

- Length, Hydrophobicity, Net Charge, Positive Ratio, Negative Ratio, and Cysteine Ratio.

2. ESM-2 Transformer Embeddings

- Leveraged a pre-trained Protein Language Model (ESM-2).
- Dimension: 320-dimensional latent space representation.

ML Models

1. Traditional Models:

- K-Nearest Neighbors (KNN): Distance-based classification.
- Linear SVM: Establishing linear decision boundaries.
- Non-Linear SVM (RBF Kernel): Capturing non-linear interactions in peptide space.
- Random Forest (RF): Ensemble of decision trees for robust feature importance.
- XGBoost (Physical): Gradient boosting applied to the 6 physicochemical descriptors.

2. ML with Transformer Embeddings:

- XGBoost + ESM-2: Integrating high-dimensional transformer embeddings with advanced gradient boosting for maximum predictive power.

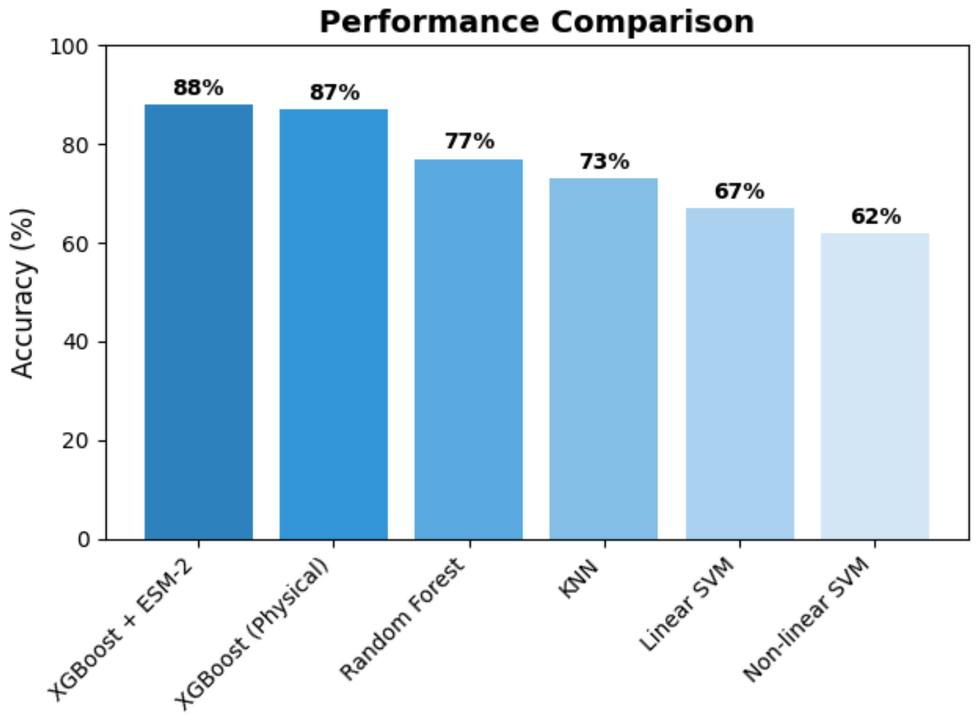
Toxicity prediction

1. Netcharge = count(K,R,H) – count(D,E) / Sequence length
2. Hydrophobicity = sum (Kyte-Doolittle value for each aminoacid) / Sequence Length
3. Aromatic % = Count of (F, W, Y) / Sequence Length

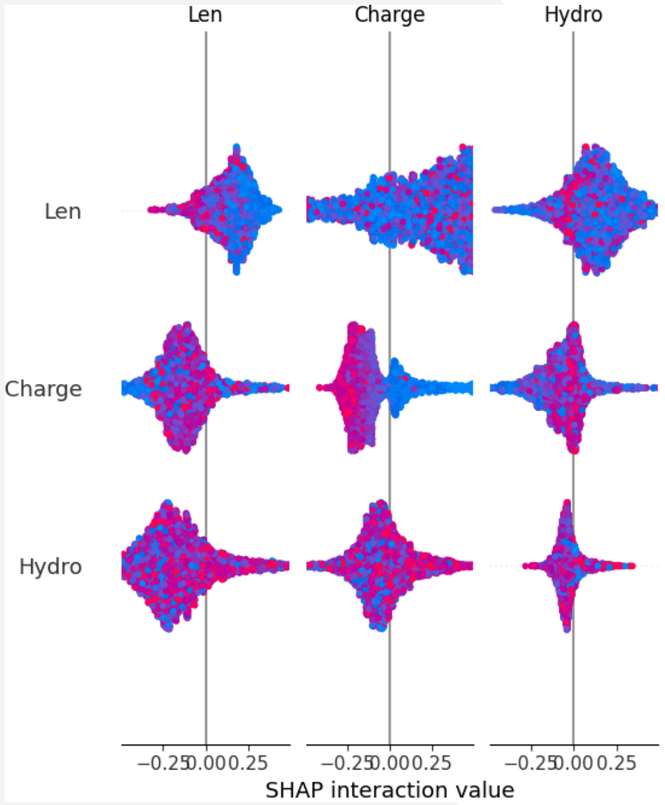
- Net Charge > 0.4
- Hydrophobicity > 0.5
- Aromatic Content > 0.1
- If all three of these conditions are met - "High Toxicity".
- If any one is not met - "Low Toxicity".

Antimicrobial activity prediction and toxicity score for a Sample AMP

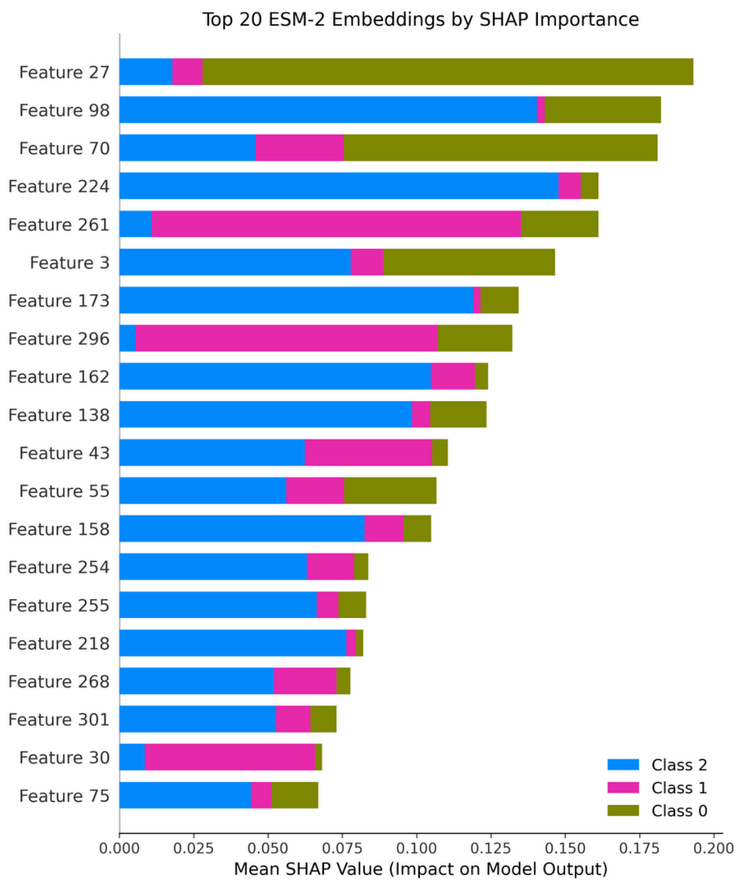
```
Enter Peptide Sequence: GIGKFLHSAKKFGKAFVGEIMNS

==================================================
PEPTIDE ANALYSIS: GIGKFLHSAKKFGKAFVGEIMNS
==================================================
PREDICTED CLASS : ANTIBACTERIAL (96.63% Confidence)
TOXICITY STATUS : Low Toxicity
--------------------------------------------------
BIO-PHYSICAL STATS:
 > Normalized Net Charge: 0.17
 > Hydrophobicity Score:  0.08
 > Aromatic Percentage:  13.0%
==================================================
```



SHAP Analysis



SHAP plot for features in the traditional models



SHAP plot for features in ESM-2 + XGBoost model

Results

Model Architecture	Features Used	Accuracy (%)	ROC-AUC	Precision	Recall (Sensitivity)	F1-Score
XGBoost + ESM-2	320 Embeddings	**882%**	**94**	88	87	87
XGBoost	6 Physico-chemical	871%	92	86	86	86
Random Forest	6 Physico-chemical	774%	85	77	76	76
KNN	6 Physico-chemical	730%	79	72	73	72
SVM (Linear)	6 Physico-chemical	675%	72	66	67	66
SVM (Non-Linear/RBF)	6 Physico-chemical	621%	68	61	62	61

The XGBoost on ESM-2 model achieved the highest overall performance with an accuracy of 88% and an ROC-AUC of 0.94.
The results were compared with confusion matrix, ROC-AUC, PR curve, Accuracy

Novelty

- Comparative evaluation of multiple ML models on the same AMP dataset
- Simultaneous prediction of activity type and toxicity
- Dataset filtering and preprocessing to reduce class imbalance
- Comparison between the usage of physicochemical and transformer based ESM-2 embeddings
- Predictions explained with SHAP plots

THANK YOU